A Bayesian Analysis of the Marth-Fox Matchup in Super Smash Bros. Melee

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Chapter 1

Introduction and Motivation

1.1 Introduction

In Super Smash Bros. Melee (or colloquially known as just “Melee”) for the Nintendo GameCube, there are 26 characters to choose from, including fan-favorites like Mario and Luigi, to less known characters like Ness and Ice Climbers. In competitive Melee, each player has four lives, also known as stocks. A player wins by depleting all of their opponents stocks before losing theirs.

Typically, in competitive games or sports, each individual or team has equal access to the same moves and techniques. For example, in chess, each player has the same number and types of pieces. In game theory, this is an example of a symmetric game, in which the players have the same strategy set [1] p. 186. However, in fighting games, each character has a unique moveset that gives them strengths and weaknesses. This is an example of an asymmetric game, in which each character has a unique strategy and set of options that can be employed. Certain characters have swords or projectiles, allowing them to attack opponents from a distance. Some characters are fast, but easy to knock off the stage, while others are slow, but more resilient to being defeated.

Additionally, Melee is unique from other fighting games, in that it is what is known as a platform fighting game. Typically, fighting games are played on a two-dimensional plane with no frills. The goal is to deplete the health of your opponent. In Melee, the goal is to knock your opponent off of the stage to deplete their stocks. Each stage has unique properties that can affect how certain characters play. Characteristics such as platforms, stage transformations, and other interactable elements affect how players approach matchups and can directly affect gameplay. It is agreed upon in the community that certain stages are better for certain characters due to the tools available to the character.

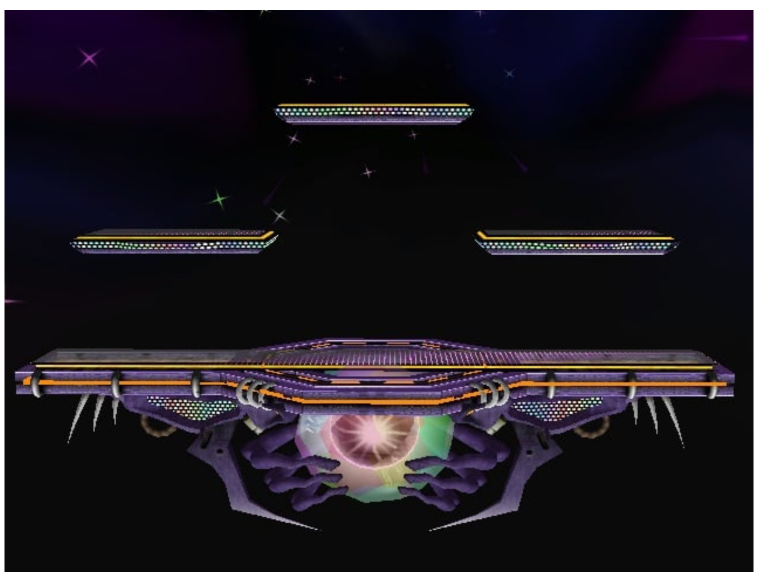

Figure 1.1: Battlefield, a stage that offers 3 platforms to extend or escape from combos.

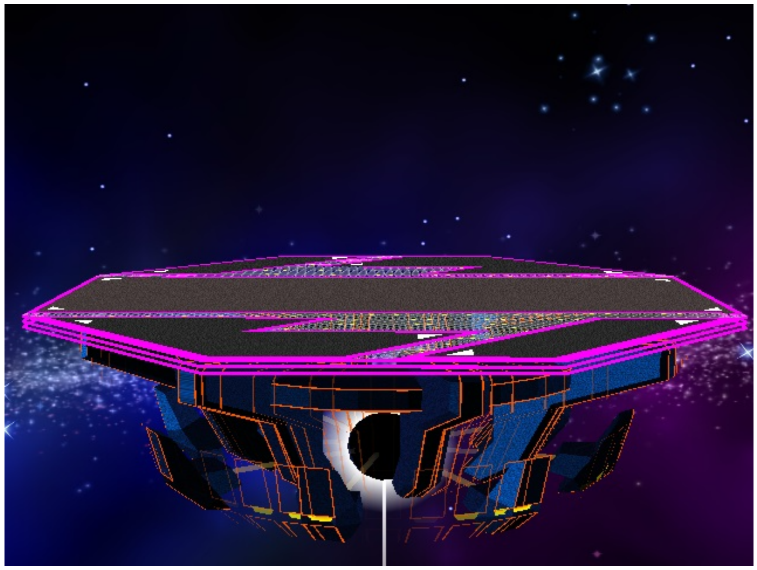

Figure 1.2: Final Destination, a stage without platforms.

Because of the asymmetric nature of Melee's characters, it is natural to assume that some characters perform better against others due to having a superior toolkit in a given matchup. To express how strong a character is, players will create a "matchup chart" or "matchup spread" for the character. This is a measurement of a character's probability of winning against other characters in the roster. When designing a matchup chart, it is typically assumed that the players of each character are of equal skill level. Matchup charts are also typically designed based on the perception of how character perform against one another, given their respective toolkits and performance on certain stages. There is typically little to no formal statistical analysis when designing these spreads.

While some matchup charts are expressed using +/- 1, 2,$\ldots$, this is not as descriptive as to how favorable or unfavorable a matchup may be. We can also use explicit probabilities when describing a matchup. Formally, let us define a matchup of character A against character B. When discussing this matchup, we would say "Charcter A's matchup against Character B is x/y," where x is the probability that character A would win, y is the probability that character B would win, and $x + y = 100$. For example, an even matchup would be designated as "50/50". A slightly favorable matchup would be "55/45", while a heavily unfavorable matchup may be expressed as "35/65".

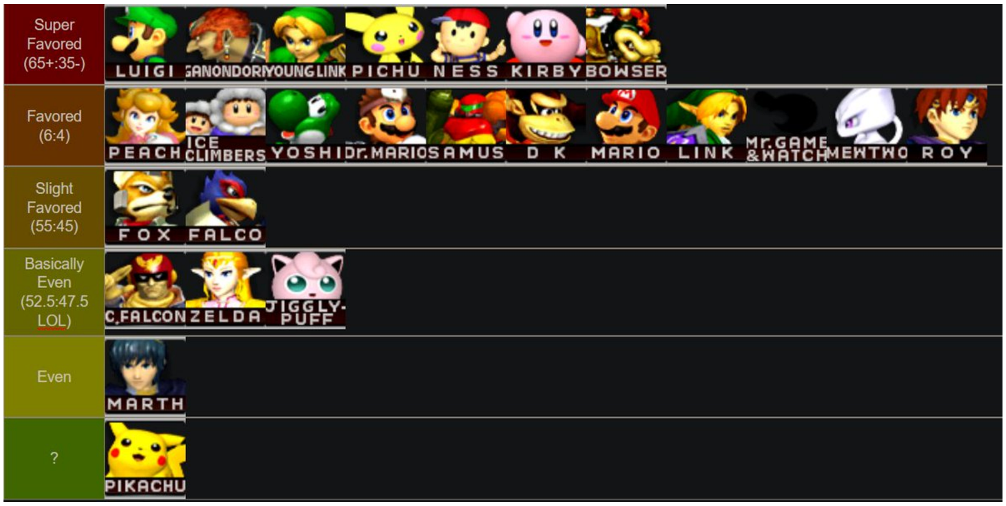

Figure 1.3: A Marth matchup chart, made by top Marth player KoDoRiN

1.2 Motivation

A tier list is an ordered list of characters, where each character is ranked based on how strong they are relative to the rest of the cast of characters. Tier lists are often subjective, and they will change over time as the strategies for each character develops over time.

One of the most prominent matchups in competitive melee is Marth vs. Fox. Both of these characters have sat near the top of almost every competitive tier list for over two decades. Both characters have been extremely successful, and many players have piloted the characters to major tournament victories. They each have strong toolkits, and the ability to perform well against any character in the game. However, there has been debate in the Melee community about how Fox and Marth perform against each other.

Most players would argue that the matchup is even. Each charcter has tools in their moveset to exploit the other character's weaknesses. However, top Melee players have expressed different opinions. The most infamous claim was made by the player William "Leffen" Hjelte. Leffen, a Fox player, argued that Marth is favored in the matchup 60/40, and it became somewhat of a joke in the Melee community. When referencing the matchup for Marth against Fox, he claimed "6-4 is probably [right], both according to results and ... Marth [having] a way easier time improving in that matchup" [2]. Although Leffen is arguably one of the best Fox players of all time and has incredible knowledge about Melee, many claimed that such a matchup spread was increadibly far-fetched.

I thought that this debate would be interesting to approach from a Bayesian lens. Given expert prior beliefs on the matchup, we could use results from individual matches to determine a more statistically sound matchup spread for a character. Furthermore, Melee has many characters that get less usage due to being considered weaker than most choices. Thus, there is little data for some of these more obscure matchups. Bayesian methods thrive in this environment, allowing us to define such matchup spreads for less-used characters.

Chapter 2

Data Collection

2.1 Slippi

Melee was originally released in 2001, with limited features. Slippi, released in June 2020, is a project designed to bring features of modern-day e-sports to Melee. These include portable replay files, complex gameplay stats, online matchmaking with improved online netcode, and more [3]. Not only did this make Melee more accessible through the introduction of online play, a feature missing from the original game, but Slippi also ushered in an era of data analytics surrounding Melee. Using Slippi game data, it is possible to easily analyze match data without having to manually record match results.

For the data used in this project, it was sourced from Chart.slp, a platform that hosted a large corpus of Slippi match data. Although it is no longer maintained, it is possible to download a backup of the database, containing data for 5,594,571 different matches. This provides sufficient data for Bayesian analysis across nearly every matchup in Melee.

To get this data into a usable format, I developed a few Python scripts to groom the data. First, for this analysis, we only care about matchup data. The information in the database file used for match replay is not needed and consumes are large amount of space. We start by parsing just match data, such as characters, stage, winner, etc.

After parsing the file for just match data, another script is used to exclude incomplete matches. Slippi allows players to disconnect from a match early, which does not result in an explicit winner. Thus, these matches are filtered out. After this filter is applied, other scripts can be run to get matches for two specific characters, and data about the stage and winner can be exported into a CSV format for analysis.

Chapter 3

Modeling the Marth-Fox Matchup

3.1 Model 1: A Simple Binomial-Beta Conjugate

For the first model, we will approach it completely analytically. We will model the win rate of a character in a matchup, without factoring in stage choice. Let θ represent the probability of a character winning a given matchup. Let the likelihood of our model be represented as a binomial distribution $x_i|\theta \sim Bin(n, \theta)$, where n represents the number of matches in the sample. A win for the character would be recorded as a 1 (success), while the loss would be recorded as a 0 (failure). We can specify a Beta prior generally as $\theta \sim Be(\alpha, \beta)$. A Beta prior is flexible, as it allows us to experiment with different initial beliefs for a given matchup.

The Beta prior can form a conjugate distribution with the posterior with respect to the Binomial likelihood. This enables us to find a fully closed-form posterior. Using existing conjugacy tables [4] p. 341. This gives the posterior $\theta|X \sim Be(\alpha + x, \beta + n - x)$. The mean of the posterior, $\frac{\alpha+x}{(\alpha+x)+(\beta+n-x)}$, gives us the expected win rate for a character in a given matchup.

For the Marth-Fox matchup, assume that we are modeling Marth's win rate against Fox. Below are graphs representing the posterior matchup distribution, given different priors.

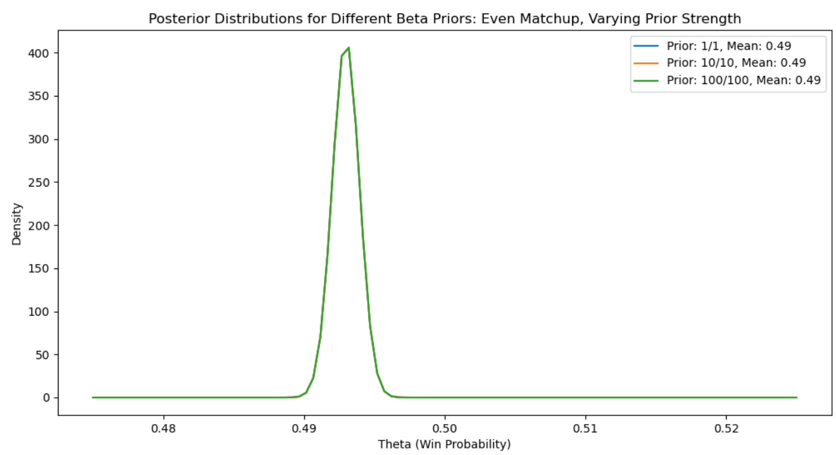

Figure 3.1: Modeling a Marth-Fox even (50/50) matchup prior with varying strengths.

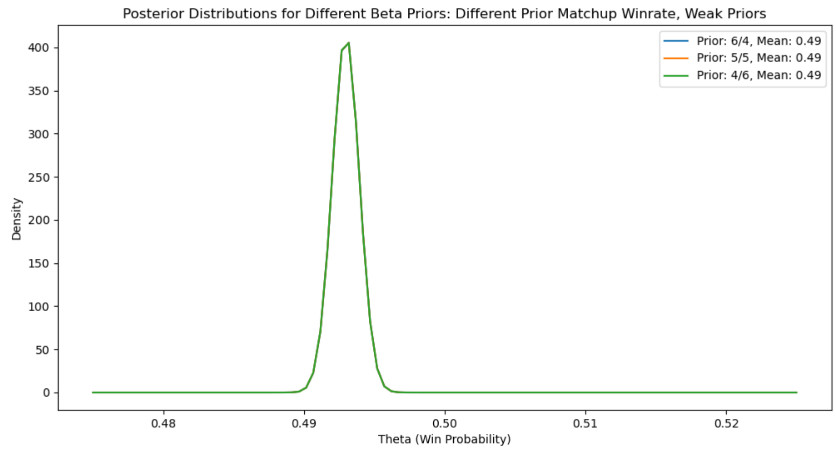

Figure 3.2: Modeling a different Marth-Fox expected win rates with weak priors.

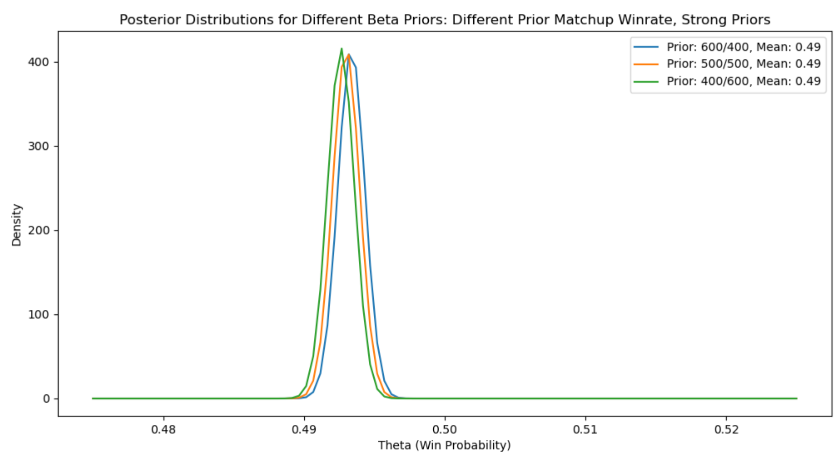

Figure 3.3: Modeling a different Marth-Fox expected win rates with strong priors.

As can be seen in these graphs, the expected win rate of Marth against Fox is greatly impacted by the observed. This makes sense because the posterior distribution parameters will be greatly influenced by the likelihood. Thus, in the case of a large data sample like this, the impact of the priors, regardless of perceived strength, is rendered nearly meaningless due to the sheer amount of observed data there is. However, for cases of uncommon matchups, where there may not be a lot of available data, the Bayesian model will allow our prior to be incorporated into the posterior to a greater degree. Below are graphs of the posterior when in a hypothetical sample of 10 Marth-Fox matches, Marth wins 8 of them.

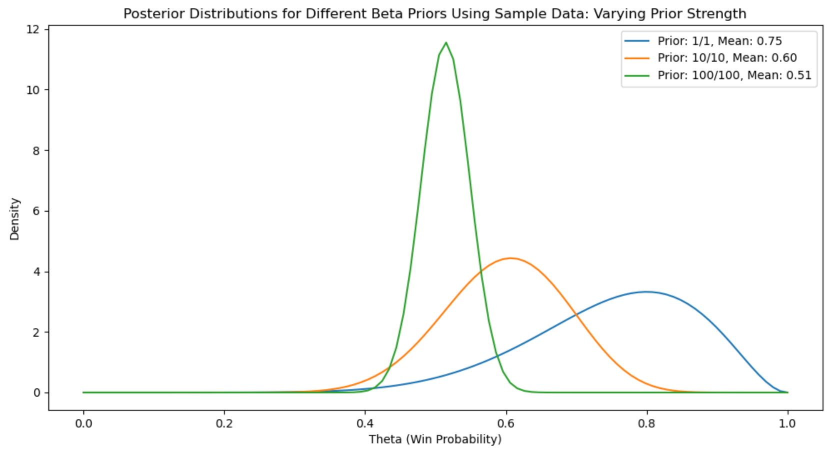

Figure 3.4: Modeling a different Marth-Fox expected win rates given an even (50/50) matchup prior using sample data.

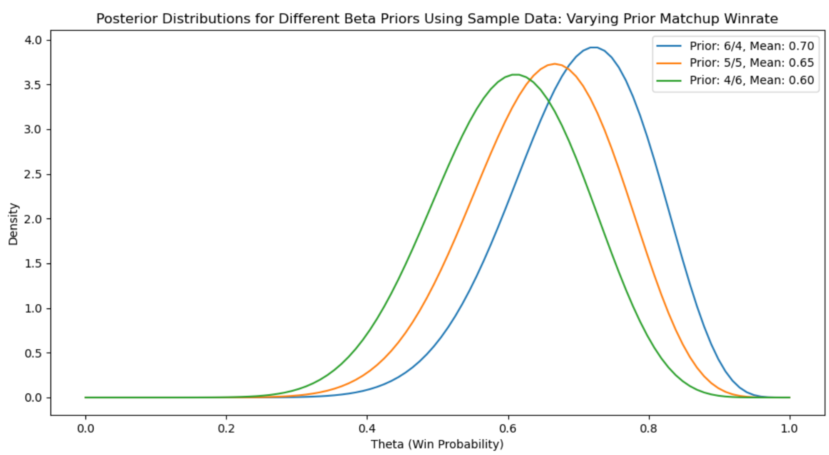

Figure 3.5: Modeling a different Marth-Fox expected win rates given varying matchup priors of similar stregth using sample data.

Looking at the above figures, it is more evident how changing a prior's strength or shape will affect the posterior distribution. Again, in the case of Melee, given $n = 26$ characters, there are $\frac{n(n+1)}{2} = \frac{26 \cdot 27}{2} = 351$ total unique matchups. By leveraging a Bayesian approach, we can more accurately model matchup probabilities for characters with significantly lower use rate in competitive settings.

Although this is a good exercise and a simple way to use Bayesian methods to better model matchups in Melee, there are several components not included in this initial model. First, as previously explained, stages in Melee often make a difference in a matchup. In the Marth-Fox matchup, it is argued that Final Destination is a strong choice for Marth due to him gaining access to techniques not available on stages with platforms. Thus, we are interested in also understanding character matchups by stage. Second, it is likely that not all of the observed matches were against evenly matched players. We may want to design a model that factors in player skill when trying to determine the true matchup spread.

3.2 Model 2: Marth-Fox Matchup by Stage

In competitive Melee, there are 6 tournament-legal stages. These are Battlefield, Final Destination, Dream Land, Fountain of Dreams, Pokemon Stadium, and Yoshi's Story. We are interested in understanding how the Marth-Fox matchup varies by stage. In competitive Melee, players will go through a process known as "stage striking", in which players take turns removing a stage from being selected until there is only one remaining [5]. It is important for a player to understand how their character performs on each stage in a given matchup when deciding which stages to strike.

To determine if there is a significant effect a stage has on the Marth-Fox matchup, we can utilize a logistic regression model. This fits since the response data (a match win or loss) is binary in nature. We define our model in the following way:

$$\begin{aligned} y_i|p_i &\sim \text{Bernoulli}(p_i), i = 1, \ldots, 6 \\ \text{logit}(p_i) &= \alpha + \gamma_i y_i \\ \alpha &\sim N(0, 10) \\ \gamma_i &\sim N(0, 10) \\ \sum \gamma_i &= 0 \end{aligned}$$

Where α represents the base "matchup effect" and γ_i represents the "stage effect" on the matchup probability. A positive value for these values indicates a favorable matchup for Marth, either generally, or given a particular stage. As a note, to speed up the model, only a subset of the full corpus of data was used (10,000 matches), which contributes to the difference in values than may have been calculated in the previous section. Below are the summary statistics for this model using a 95% HDI credible interval:

	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
alpha	-0.284	0.020	-0.321	-0.243	0.000	0.000	5294.000	3306.000	1.000
gamma[YOSHIS_STORY]	0.121	0.045	0.036	0.210	0.001	0.001	5943.000	2928.000	1.000
gamma[FOUNTAIN_OF_DREAMS]	-0.019	0.046	-0.107	0.070	0.001	0.001	5386.000	3267.000	1.000
gamma[FINAL_DESTINATION]	0.069	0.046	-0.019	0.163	0.001	0.001	6276.000	2805.000	1.000
gamma[POKEMON_STADIUM]	-0.081	0.043	-0.162	0.004	0.001	0.001	5336.000	2909.000	1.000
gamma[BATTLEFIELD]	-0.019	0.044	-0.108	0.060	0.001	0.001	6070.000	3172.000	1.000
gamma[DREAMLAND]	-0.070	0.046	-0.161	0.017	0.001	0.001	5253.000	3418.000	1.000

Looking at this data, we see that the credible for α does not cross 0. This implies that, disregarding any additional stage factor, Marth is likely to have a losing matchup against Fox. Specifically, when converting the log-odds to a probability, we estimate the base probability of Marth winning against Fox to be approximately 42.9%. This contradicts both the assumption that the matchup is even, or the possibility that the matchup is in Marth's favor.

Analyzing the γ_i values, we can analyze what stages could be beneficial for Marth. Yoshi's Story appears to be the only stage where we can conclusively determine (given the 95% HDI credible interval) that it is more likely that Marth will win on this stage. While not conclusive, Final Destination appears to be in Marth's favor. Similarly, Dream Land and Pokemon Stadium appear to lean in Fox's favor, although this is also not conclusive. Battlefield and Fountain of Dreams appear to be relatively even for both characters. Given this data, a Marth player's choice for stage bans or selections can be backed by concrete statistics.

We also can conduct a hypothesis test to see if the belief that the matchup being 60/40 in Marth's favor holds any merit. We can do this by checking if invlogit$(\alpha) >= 0.6$ for each sampled α in the chain. When performing this test, we get the following output:

	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
h1	0.000	0.000	0.000	0.000	0.000	NaN	4000.000	4000.000	NaN

It can be seen that in the sampling there were no samples that fit this criteria. Thus, this is a strong indication that the belief that the Marth Fox matchup is 60/40 in Marth's favor is not supported at all by this model.

Note: When conducting the hypothesis test, arviz returned runtime warnings due to attempting to divide by zero. This is likely due to the hypothesis test returning no viable samples given the constraint invlogit$\alpha >= 0.6$. *These warnings are safe to ignore in this case.*

The downside of this model is that it does not factor in player skill. The benefits of certain stages are emphasized more when a player has the necessary skill to exploit certain techniques. Additionally, it is possible that the matchup winrate in general may shift depending on the skill level of players. The next model will propose an approach to incorporate Slippi ratings into our model.

3.3 Model 3: Factoring in Elo

In newer versions of Slippi, a ranked mode was added. In this mode, players are given a rating. A higher number is indicative of a more skilled player, and vice versa. We can use this information as a factor in modeling the matchups in Melee.

Slippi follows a rating system in which players are categorized into different "ranks" based on their specific rating [6]. For instance, if a player has a rating of 1450, they would have a rank of Gold 1. This distinction is important in how Elo is factored into the model.

Rank	ELO Required
Bronze 1	0
Bronze 2	766
Bronze 3	914
Silver 1	1055
Silver 2	1189
Silver 3	1316
Gold 1	1436
Gold 2	1549
Gold 3	1654
Plat 1	1752
Plat 2	1843
Plat 3	1928
Dia 1	2004
Dia 2	2074
Dia 3	2137
Master 1	2192
Master 2	2275
Master 3	2350
Grandmaster	Top of Region AND > 2192

Figure 3.6: Rating cutoffs for each rank in Slippi.

The Bradley-Terry model provides an approach in which we can model the probability of Marth player winning a match against Fox player given the strength of the players. Let β_i, β_j represent the perceived "strength" of the Marth and Fox player respectively, in terms of Elo. The log-odds of the Marth player beating the Fox player is equal to the difference of β_i and β_j[7]. Through strategic ordering of parameters, we can also introduce other effects that can factor into the outcome of a game between two players, including the previously explored "matchup" factor and "stage" factor. Thus, we can naturally extend the previously defined model for Marth's matchup against Fox in a vacuum, to now model the probability of a specific Marth player winning against a Fox player.

Unfortunately, the data sampled for this project did not have access to player Elo data. To prove out this new model, Elo data was synthesized based on the existing range of Slippi ratings. To normalize the Elo data into relative skill differences, I opted to divide the difference in player elo by 300. This aligns with the approximate elo difference between major ranks (i.e. between Silver and Gold, Gold and Platinum, etc.) across Slippi's ranked system. Thus, for the modified Bradley-Terry model, we use a scale factor of 300 and a base of e, slightly different than the traditional setup for elo in chess, which utilizes a scale factor of 400 and base of 10 [8].

More formally, we can define our new model in the following way:

$$
\begin{aligned}
y_i|p_i &\sim \text{Bernoulli}(p_i), i = 1, \ldots, 6 \\
\text{logit}(p_i) &= \beta(\frac{\epsilon_m - \epsilon_f}{300}) + \alpha + \gamma_i y_i \\
\alpha &\sim N(0, 10) \\
\beta &\sim N(0, 10) \\
\gamma_i &\sim N(0, 10) \\
\sum \gamma_i &= 0
\end{aligned}
$$

Where α and γ_i are the same as previously defined, ϵ_m, ϵ_f is the Slippi rating of the Marth and Fox player respectively, and β represents the effect elo plays into the matchup. Running this new model once more using the same data produces the following results:

	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
alpha	-0.284	0.020	-0.322	-0.244	0.000	0.000	6584.000	2632.000	1.000
beta	0.002	0.010	-0.018	0.022	0.000	0.000	6185.000	2853.000	1.000
gamma[YOSHIS_STORY]	0.120	0.042	0.041	0.203	0.001	0.001	5576.000	2730.000	1.010
gamma[FOUNTAIN_OF_DREAMS]	-0.018	0.047	-0.107	0.073	0.001	0.001	7786.000	3162.000	1.000
gamma[FINAL_DESTINATION]	0.068	0.047	-0.023	0.156	0.001	0.001	7329.000	2771.000	1.000
gamma[POKEMON_STADIUM]	-0.081	0.044	-0.172	0.001	0.001	0.001	6830.000	2919.000	1.000
gamma[BATTLEFIELD]	-0.019	0.043	-0.106	0.065	0.001	0.001	7269.000	3220.000	1.000
gamma[DREAMLAND]	-0.070	0.047	-0.156	0.023	0.001	0.001	7406.000	3410.000	1.000

Looking at the data for β, it appears that a difference in Slippi rating has little impact on the projected matchup. This is almost certainly due to the elo data being randomly generated; in fact, it likely serves as an example to show that the model is recognizing rating as a "random" factor of sorts. Given real, live data, I would expect this parameter to have the potential of changing.

For the α and γ_i parameters, little has changed, also due to β having little impact on the matchup. While the matchup factor would likely remain the same, I would expect the stage effects to potentially diverge as skill difference is accounted for.

Chapter 4

Conclusion

4.1 Conclusion

Using the Binomial-Beta model with the Slippi data and varying priors, we determined that the Marth-Fox matchup is close enough to be considered even. This contrasts Leffen's claim that the Marth-Fox matchup is heavily favored for Marth. With the various logistic regression models, using a subset of the Slippi data, we determined that Yoshi's Story is definitively a strong stage for Marth against Fox. While not definitive, Final Destination appears to be in Marth's favor, while Dream Land and Pokemon Stadium appear to lean in Fox's favor. Battlefield and Fountain of Dreams appear to be relatively even. Final Destination is often regarded as Marth's best stage in the matchup, so it was interesting to see the average Marth player perform better on Yoshi's Story, relatively speaking. Finally, we adapted the logistic regression model to factor in the relative difference of player Slippi ratings. Although this data was not in our dataset, future Slippi match data containing player ratings could use such data in this model for analysis.

There are nearly infinite variables when it comes to fighting games. A matchup may be awful on paper, but a player may practice it to such a degree that they make seem as though their character wins a matchup. Additionally, in competitive Melee, no two games are truly independent. Players use information about their opponents playstyles, habits, and skills to make strategic, on-the-fly adaptations. The free-flowing nature of a game like Melee results in infinite possibilities, and it is a main contributer to why its competitive scene has not only survived, but thrived over the last 25 years.

For the scope of this project, I opted to make assumptions about match independence and select a few major matchup factors to simplify the creation of a logistic regression model. There are several steps to improve upon the existing model. First, the model could utilize actual tournament data, where we could identify win probabilities given the result of the current match count in a best-of-five set. Additionally, data could be polled for specific Marth players to see how they do against specific Fox players or Fox players in general. A similar approach could be used for other characters in the cast. We could also run the model using data from only top players to determine if matchups change at different skill levels, or we could introduce an interaction effect to see if a combination of skill difference and stage selection work together to affect a matchup. Analysis of data like this could improve the viewer experience by showing live probabilities of a player winning given the current state of a set, or can be used to predict winners of major tournaments. Like Melee itself, there are nearly infinite possible ways to leverage Baysian methods to uncover something new about a beloved game.

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